

$$\begin{array}{c} ? \\ ? \\ ? \\ ?? \\ ?? \\ ?? \\ ? \\ ? \\ ? \\ ? \\ ? \\ x_ix_j \\ ? \\ ? \\ ? \\ ? \\ ? \\ C_1 \\ C_2 \\ w \end{array}$$

$$(1) \quad y = \mathbf{w}^T x$$

$$\begin{array}{c} S_W \\ S_B \end{array}$$

$$(2) \quad \mathbf{S}_W = \sum_{n \in C_1} (\mathbf{x}_n - \mathbf{m}_1)(\mathbf{x}_n - \mathbf{m}_1)^T + \sum_{n \in C_2} (\mathbf{x}_n - \mathbf{m}_2)(\mathbf{x}_n - \mathbf{m}_2)^T$$

$$(3) \quad J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}$$

$$?$$

$$(4) \quad \mathbf{w}^* \propto \mathbf{S}_W^{-1}(\mathbf{m}_1 - \mathbf{m}_2)$$

$$(5) \quad \mathbf{m} = \frac{1}{N} \sum_{k=1}^K N_k \mathbf{m}_k, \mathbf{S}_W = \sum_{k=1}^K \mathbf{S}_k, \mathbf{S}_B = \sum_{k=1}^K N_k (\mathbf{m}_k - \mathbf{m})(\mathbf{m}_k - \mathbf{m})^T$$

$$?$$

$$(6) \quad J(\mathbf{W}) = \frac{|\mathbf{W}^T \mathbf{S}_B \mathbf{W}|}{|\mathbf{W}^T \mathbf{S}_W \mathbf{W}|}$$

$$?$$

$$(7) \quad \mathbf{S}_B \mathbf{w}_i = \lambda_i \mathbf{S}_W \mathbf{w}_i$$

$$\begin{array}{c} C \\ C- \\ | \\ ? \\ ? \\ ? \\ ? \\ ? \end{array}$$

$$(8) \quad \delta_k(x) = -\frac{1}{2} \ln |\Sigma_k| - \frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) + \ln \pi_k$$

$$\begin{array}{c} p > \\ n \\ \mathbf{S}_W \\ \mathbf{S}_W^{-1} \\ \mathbf{S}_W^{-1} \\ ? \\ ? \\ \mathbf{W} \end{array}$$

$$(9) \quad \mathbf{W}^T \mathbf{S}_W \mathbf{W} = \mathbf{I}$$

$$(10) \quad \mathbf{S}_W = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T,$$

$$\begin{array}{c} \mathbf{U} \\ \mathbf{\Lambda} \\ \mathbf{W} = \\ \mathbf{U} \mathbf{\Lambda}^{-1/2} \\ \mathbf{W} \\ \mathbf{S}_W \\ \mathbf{S}_W \\ \mathbf{X} \\ \mathbf{X} = \\ \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T \\ \sum \\ \sigma_i \\ \tau \end{array}$$